ITU-MiniTwit

A Report on Design, Operation, and Evolution

Group X

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BSc DevOps, Software Evolution and Software Maintenance
IT University of Copenhagen — Spring 2026

05/05/2026

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1 Introduction

Author: Member A

One paragraph framing the project, the team's high-level goals, and pointers to: the main repository, issue tracker, monitoring dashboard, and logging dashboard. 100 words.

2 System's Perspective

2.1 Architecture and Design

Author: Member A

Describe the architecture using three viewpoints. If you invent a notation, include a legend.

2.2 Dependencies

Author: Member B

Layer	Tool / Technology	Purpose
Runtime	Bun + Elysia	HTTP server, request handling
Data	PostgreSQL @ PlanetScale	Primary persistence
Infra	Docker Swarm on Hetzner+OCI	Hosting & orchestration
CI/CD	GitHub Actions	Build, test, deploy
Observability	Grafana / Loki / Prometheus	Metrics & logs
Secrets	Doppler	Secret distribution

Table 1: Key dependencies, by layer.

2.3 Current State

Author: Member B

Static analysis findings (Semgrep / ESLint / Sonar), test coverage, technical-debt hotspots, and quality gates enforced in CI. Cite concrete numbers.

3 Process Perspective

3.1 CI/CD Pipeline

Author: Member C

End-to-end stages and tools, including triggers, gates, and artifacts produced.

3.2 Deployment and Release

Author: Member C

Versioning scheme, environments, rollout strategy, rollback. Link a representative release.

3.3 Availability and Scaling

Author: Member C

Replicas, autoscaling, multi-region setup, failure modes, recovery procedures.

3.4 Monitoring

Author: Member D

What is monitored (golden signals + business metrics), collection mechanism, dashboards. Link the dashboards.

3.5 Logging

Author: Member D

What is logged, structured-log conventions, aggregation, retention. Link the logging UI.

3.6 Security Hardening

Author: Member D

Threat-model summary and concrete mitigations: TLS, secrets management, dependency scanning, container hardening, branch protection, SAST/DAST. Reference any incidents handled.

4 Reflection Perspective

Author: Member E

Biggest issues with respect to **evolution & refactoring**, **operation**, and **maintenance** — each anchored to specific commits, PRs, and issues. Close with a paragraph on the team's "DevOps style".

5 Use of Generative AI

Author: Member E

Tools used, tasks they were applied to, how they were used, and a brief reflection on whether they helped or hindered. Follow ITU's GenAI guidelines.